

Lecture 13: Generalized Linear Models

One Framework · Three Components · Many Models

Previously, on Lecture 12...

OLS: $Y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$. LINE assumptions. t -tests for coefficients.

Gauss–Markov: OLS is BLUE. Best among linear unbiased estimators.

Diagnostics: Residual plots, QQ plots, leverage, Cook's distance.

Logistic: Binary Y . MLE (no closed form). Z -test. Odds ratios = $e^{\hat{\beta}_j}$.

Both: Fit model $\rightarrow$ test coefficients $\rightarrow$ check overall fit $\rightarrow$ diagnostics.

Today: OLS and logistic regression look different — but they are **the same machine** with different settings. That machine is the GLM.

Part I: Why Do We Need GLMs?

Not all data is Normal. Not all relationships are linear.

Three Problems, Three Response Types

Continuous

House prices
Blood pressure
Temperature

$$Y \in \mathbb{R}$$

$Y \sim \text{Normal}$

OLS regression

Binary

Spam or not
Admit or reject
Click or skip

$$Y \in \{0, 1\}$$

$Y \sim \text{Bernoulli}$

Logistic regression

Counts

Website visits
Goals per match
Insurance claims

$$Y \in \{0, 1, 2, \dots\}$$

$Y \sim \text{Poisson}$

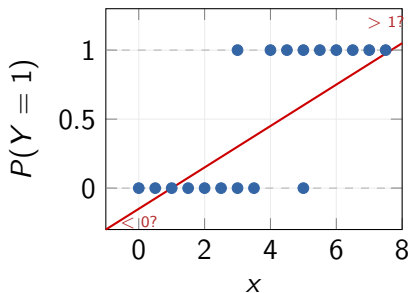
Poisson regression

All three are **special cases of one framework**: the Generalized Linear Model.

Same estimation (MLE), same inference (Z-tests, CIs), same diagnostics.

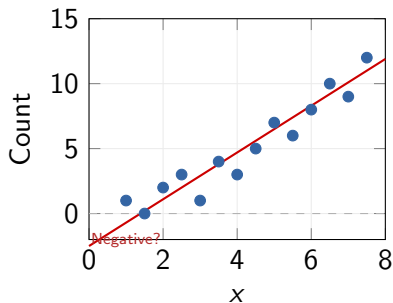
Why Not Just Use OLS for Everything?

OLS on binary data



Binary: OLS predicts $P(Y=1)$ outside $[0, 1]$. Nonsensical. Variance is not constant.

OLS on count data

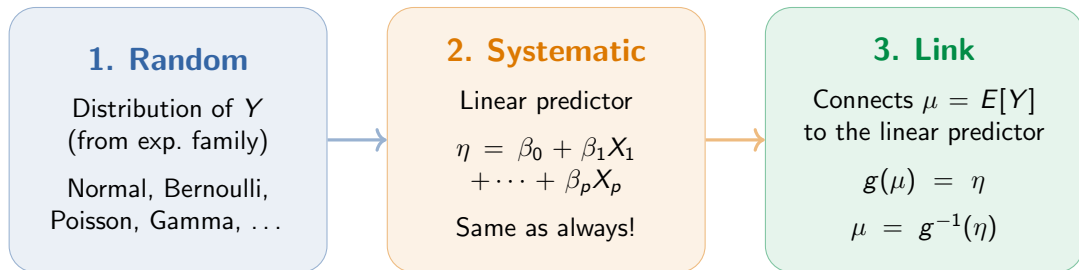


Counts: OLS predicts negative counts. Variance grows with mean. Residuals are skewed.

Part II: The Three Components

Random + Systematic + Link = GLM

The Three Components of a GLM



Key insight: We don't model Y directly as linear in X .
Instead, we model some **transformation** $g(\mu)$ as linear in X .
The link function g ensures predictions make sense for the data type.

Exponential Family: The Common Thread

Connects to L3 (exponential family) and L5 (MLE)

A distribution is in the **exponential family** if its PDF/PMF can be written as:

$$f(y \mid \theta) = h(y) \cdot \exp(\eta(\theta) T(y) - A(\theta))$$

$\eta(\theta)$ = natural parameter, $T(y)$ = sufficient statistic, $A(\theta)$ = log-partition function.

Distribution	θ	$\eta(\theta)$	$A(\theta)$	Mean
Normal(μ , known σ^2)	μ	μ/σ^2	$\mu^2/(2\sigma^2)$	μ
Bernoulli(p)	p	$\log \frac{p}{1-p}$	$\log(1 + e^\eta)$	p
Poisson(λ)	λ	$\log \lambda$	$\lambda = e^\eta$	λ
Exponential(λ)	λ	$-\lambda$	$-\log \lambda$	$1/\lambda$

The **canonical link** sets $g(\mu) = \eta(\theta)$: the natural parameter *is* the linear predictor.

Link Functions: The Key Examples

Identity

$$g(\mu) = \mu$$

$$\mu = \eta$$

Mean *is* the
linear predictor.

Normal / OLS

Logit

$$g(\mu) = \log \frac{\mu}{1-\mu}$$

$$\mu = \frac{1}{1 + e^{-\eta}}$$

Maps $(0, 1)$ to $\mathbb{R}$.

Bernoulli / Logistic

Log

$$g(\mu) = \log \mu$$

$$\mu = e^{\eta}$$

Ensures $\mu > 0$.
Multiplicative effects.

Poisson / Counts

These are the **canonical links** — the “natural” choice for each distribution.

Other links exist (probit for binary, inverse for Gamma), but canonical is the default.

Part III: OLS Is a GLM

Everything you already know, rewritten in GLM language

Linear Regression = GLM(Normal, Identity)

Random:

$$Y \sim N(\mu, \sigma^2)$$

Systematic:

$$\eta = X\beta$$

Link: Identity

$$g(\mu) = \mu = \eta$$

$$g(\mu) = \mu = \eta = X\beta \quad \Rightarrow \quad E[Y | X] = X\beta$$

This is just OLS! The identity link means: “don’t transform anything.”

Special properties of Normal + Identity:

- MLE = OLS (minimizing RSS = maximizing Normal likelihood)
- Exact t and F distributions (not just asymptotic)
- Closed-form solution: $\hat{\beta} = (X^T X)^{-1} X^T Y$

Other GLMs need iterative algorithms (no closed form).

Part IV: Logistic Is a GLM

Same framework, different distribution and link

Logistic Regression = GLM(Bernoulli, Logit)

Random:

$$Y \sim \text{Bernoulli}(p)$$

Systematic:

$$\eta = X\beta$$

Link: Logit

$$g(p) = \log \frac{p}{1-p} = \eta$$

$$g(p) = \log \frac{p}{1-p} = X\beta \quad \Rightarrow \quad p = \frac{1}{1 + e^{-X\beta}}$$

This is exactly the logistic regression from L12. The logit link maps $(0, 1) \rightarrow \mathbb{R}$.

Properties of Bernoulli + Logit:

- No closed form — MLE via iterative algorithms (Newton–Raphson / IRLS)
- Asymptotic inference: Z-tests and Wald CIs (not exact t)
- Coefficients $\hat{\beta}_j$ are log-odds ratios: $e^{\hat{\beta}_j} = \text{odds ratio}$
- Variance is $p(1-p)$ — automatically depends on the mean (no separate σ^2)

Part V: Poisson Regression

The new member of the GLM family: modeling counts

Poisson Regression = GLM(Poisson, Log)

Random:

$$Y \sim \text{Poisson}(\lambda)$$

Systematic:

$$\eta = X\beta$$

Link: Log

$$g(\lambda) = \log \lambda = \eta$$

$$g(\lambda) = \log \lambda = X\beta \quad \Rightarrow \quad \lambda = e^{X\beta}$$

The log link ensures $\lambda > 0$ (counts can't be negative) and gives **multiplicative** effects.

Multiplicative interpretation: A 1-unit increase in X_j *multiplies* the expected count by e^{β_j} .

$e^{\beta_j} = 1.2$ means 20% more events. $e^{\beta_j} = 0.8$ means 20% fewer. This is the count analogue of the odds ratio in logistic regression.

Example: Bike Rentals

Predicting daily bike rentals from temperature and weekday

$n = 365$ days. Model: $\log(\lambda_i) = \beta_0 + \beta_1 \cdot \text{Temp}_i + \beta_2 \cdot \text{Weekend}_i$.

	Estimate	SE	Z	p-value	$e^{\hat{\beta}}$
(Intercept)	5.20	0.08	65.0	< 0.001	—
Temperature ($^{\circ}\text{C}$)	0.045	0.003	15.0	< 0.001	1.046
Weekend	0.18	0.04	4.5	< 0.001	1.20

Temperature: $e^{0.045} = 1.046$

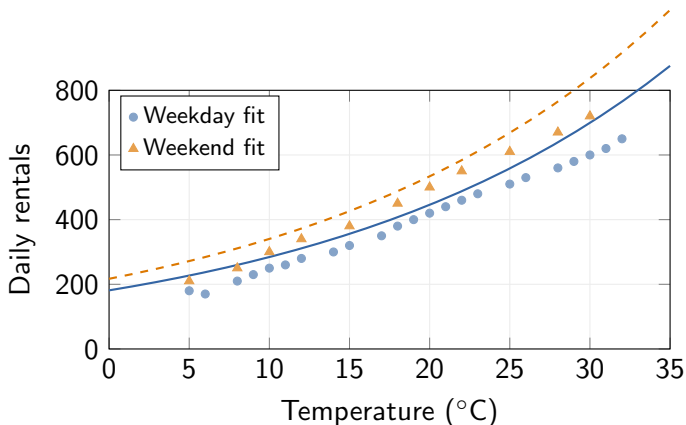
Each degree warmer $\Rightarrow$
4.6% more rentals.

Weekend: $e^{0.18} = 1.20$

Weekends have 20%
more rentals than weekdays.

At 20°C on a weekday: $\hat{\lambda} = e^{5.20+0.045 \times 20+0.18 \times 0} = e^{6.10} \approx 446$ rentals.

Poisson: Predicted vs Observed



The exponential curves naturally capture the **accelerating** relationship: more temperature = proportionally more rentals. OLS would predict a straight line.

Overdispersion: When Poisson Isn't Enough

Poisson assumes $\text{Var}(Y) = \mu$. Real count data often has $\text{Var}(Y) > \mu$ (**overdispersion**).

Poisson: $\text{Var}(Y) = \mu$

Mean and variance are tied.

If mean = 10, variance = 10.

Often too restrictive.

Negative Binomial:

$$\text{Var}(Y) = \mu + \mu^2/\theta$$

Extra parameter θ absorbs the excess variance.

Default for real count data.

Diagnostic: Fit Poisson, check residual deviance / df. If $\gg 1$: overdispersion.

Fix: Use Negative Binomial

(`statsmodels.GLM(family=NegativeBinomial())`) or
Quasi-Poisson (same $\hat{\beta}$, inflated SEs).

Part VI: How GLMs Are Fit

MLE, IRLS, deviance, and the same inference workflow

MLE for GLMs: IRLS Algorithm

Step 1: Start with initial guess $\hat{\beta}^{(0)}$ (e.g., from OLS on transformed data).

Step 2: Compute weights w_i and working response z_i from current $\hat{\beta}$.

Step 3: Solve **weighted** least squares: $\hat{\beta}^{(t+1)} = (X^\top W X)^{-1} X^\top W z$.

Step 4: Repeat until convergence. This is Newton–Raphson on the log-likelihood.

IRLS = Iteratively Reweighted Least Squares.

At each step, solve a *different* weighted regression.
Weights w_i depend on the link and current fitted values.

For Normal + Identity: converges in **one step**
(= OLS). For others: typically 5–10 iterations.

Inference: Same Workflow, Different Details

	OLS	Logistic	Poisson
Estimation	Closed form	MLE (IRLS)	MLE (IRLS)
Test stat	t_{n-p-1} (exact)	$Z \sim N(0, 1)$ (asymp.)	$Z \sim N(0, 1)$ (asymp.)
SE source	$\hat{\sigma}^2(X^\top X)^{-1}$	$\mathcal{I}(\hat{\beta})^{-1}$	$\mathcal{I}(\hat{\beta})^{-1}$
CI	$\hat{\beta}_j \pm t \cdot \text{SE}$	$\hat{\beta}_j \pm z \cdot \text{SE}$	$\hat{\beta}_j \pm z \cdot \text{SE}$
Interpret	$\Delta Y = \hat{\beta}_j$	OR = $e^{\hat{\beta}_j}$	Rate ratio = $e^{\hat{\beta}_j}$
Overall test	F-test	LRT (χ^2)	LRT (χ^2)

The pattern is always the same:

Fit model $\rightarrow$ get $\hat{\beta}_j$ and SE $\rightarrow$ test $H_0: \beta_j = 0 \rightarrow$ CI $\rightarrow$ interpret
 $\rightarrow$ diagnostics.

Only the distribution, link, and test statistic change.

Deviance: The GLM Analogue of RSS

$$D = 2 \left[\ell(\text{saturated}) - \ell(\hat{\beta}) \right]$$

Deviance = $2 \times$ (log-lik of perfect model – log-lik of fitted model).
Measures how far the fitted model is from perfectly explaining the data.

For Normal (OLS):

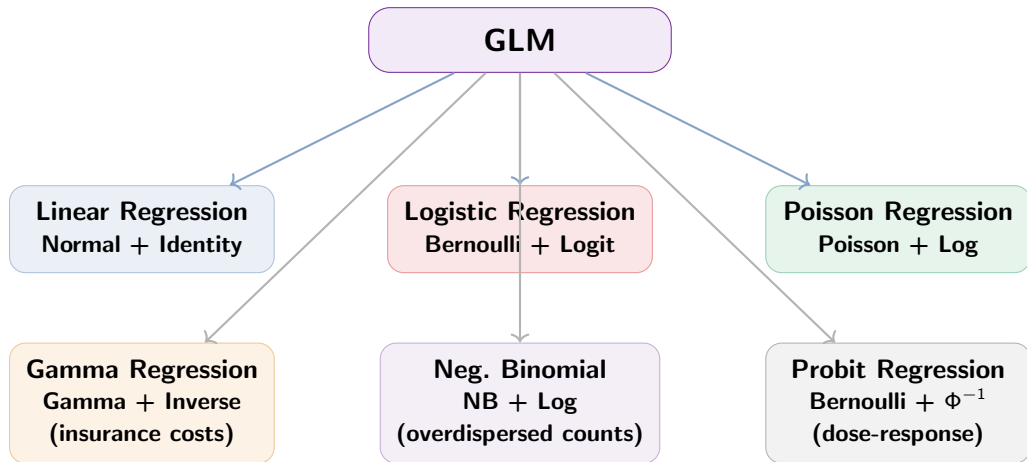
Deviance = RSS.
The familiar residual
sum of squares.

Model comparison (LRT):

$\Delta D = D_{\text{small}} - D_{\text{large}} \sim \chi^2_{\Delta p}$
Same idea as the LRT
from L10 (Wilks' theorem).

AIC = $D + 2p$ (penalize complexity). Lower AIC = better model. Replaces R^2_{adj} for GLMs.

The GLM Family Tree



All share the same three-component structure. All use MLE. All have the same inference workflow.

Choosing the Right GLM

Continuous, roughly symmetric: Normal + Identity (OLS). The classic.

Binary (yes/no, 0/1): Bernoulli + Logit (logistic regression). Interpret via odds ratios.

Counts (0, 1, 2, ...): Poisson + Log. Check for overdispersion; switch to Neg. Binomial if needed.

Positive continuous (costs, durations): Gamma + Inverse (or Log). Right-skewed, variance $\propto \mu^2$.

Proportions (0 to 1, not binary): Beta regression (not strictly a GLM, but same spirit).

Not sure? Look at the response variable. Its type determines the distribution. The link follows.

Summary: Generalized Linear Models

Three components: Random (distribution) + Systematic ($\eta = X\beta$) + Link ($g(\mu) = \eta$).

OLS: Normal + Identity. Closed form. Exact t and F .

Logistic: Bernoulli + Logit. Binary data. Odds ratios = $e^{\hat{\beta}}$.

Poisson: Poisson + Log. Count data. Rate ratios = $e^{\hat{\beta}}$. Check overdispersion.

Estimation: MLE via IRLS. OLS is the special case that converges in one step.

Inference: Z -tests and Wald CIs (asymptotic). LRT for model comparison.

Deviance: GLM analogue of RSS. $AIC = D + 2p$ for model selection.

Big picture: One framework, many models. Same workflow for all.

Practical: GLMs in Python

1. Poisson regression:

- ▶ Simulate count data: $Y \sim \text{Poisson}(e^{1+0.5X})$
- ▶ Fit with `statsmodels.GLM(family=Poisson())`
- ▶ Interpret coefficients as rate ratios. Plot fitted curve

2. Compare OLS vs Poisson:

- ▶ Fit OLS to the same count data. Compare predictions
- ▶ Which model predicts negative counts?

3. Overdispersion check:

- ▶ Simulate overdispersed counts (Neg. Binomial)
- ▶ Fit Poisson. Check deviance/df. Fit Neg. Binomial. Compare AIC

4. Logistic as GLM:

- ▶ Fit logistic regression using `GLM(family=Binomial())`
- ▶ Verify it gives the same results as Logit

Homework

1. A hospital records daily ER visits (Y) with day of week and temperature. (a) Why is Poisson regression more appropriate than OLS? (b) Fit a Poisson GLM. Interpret coefficients as rate ratios. (c) Check for overdispersion. If present, refit with Neg. Binomial. Compare AIC.
2. Show that for $\text{Normal}(\mu, \sigma^2)$ with known σ^2 : (a) The canonical link is the identity. (b) The MLE is $\hat{\mu} = \bar{Y}$ (= OLS intercept-only). (c) The deviance equals RSS.
3. A study models insurance claim amounts (positive, right-skewed). (a) Which GLM family and link would you use? Why? (b) Fit with `GLM(family=Gamma(link=Log()))`. Compare predictions with OLS.
4. **Unifying exercise:** Take one dataset and fit OLS, logistic (binarize Y), and Poisson (round Y). Compare the `.summary()` tables.

Recommended Resources

Video: StatQuest — Generalized Linear Models (GLMs) Clearly Explained

Excellent visual intro to the three components. Also: “Poisson Regression.”

Reading: Nelder & Wedderburn (1972), “Generalized Linear Models”

The original paper. Surprisingly readable. Introduced the GLM framework and IRLS.

Textbook: McCullagh & Nelder, “Generalized Linear Models” (2nd ed.)

The definitive reference. Chapters 1–4 cover theory; rest covers specific models.

Reading: James et al., “ISLR” — Ch. 4 (Logistic) + supplements

Intuitive treatment of logistic; Poisson in the online supplements.

Python: `statsmodels.genmod.generalized_linear_model.GLM`

`family=Gaussian(), Binomial(), Poisson(), Gamma(), NegativeBinomial().`

Questions?

Next: Lecture 14 — Causal inference